Datahub Modbus_TCP Protocol
V1.3

Changelog

V1.0	Only support X3-Mega G2, X3-Forth Datahub1000 FW: V014.01(Test Version)	Keven BX Li
V1.1	Integrated X3-Hybrid-G4, X3-FIT-G4, X3-Mega G2, X3-Forth, X3-Pro-G2, X3-MIC-G2 into Datahub Modbus function, you can use Datahub Modbus to control when these Inverters exist at the same system. Datahub1000 FW: V015.07(Official Version)	Keven BX Li
V1.2	Integrated X3-Ultra, X3-IES, into Datahub Modbus function. Add new function Add new address for system SOC, each device’s SOC, each device working status, Grid related Fault Alarm	Keven BX Li
V1.3	Correct the Modbus address from 40446 to 40451	Keven BX Li

Please notice:

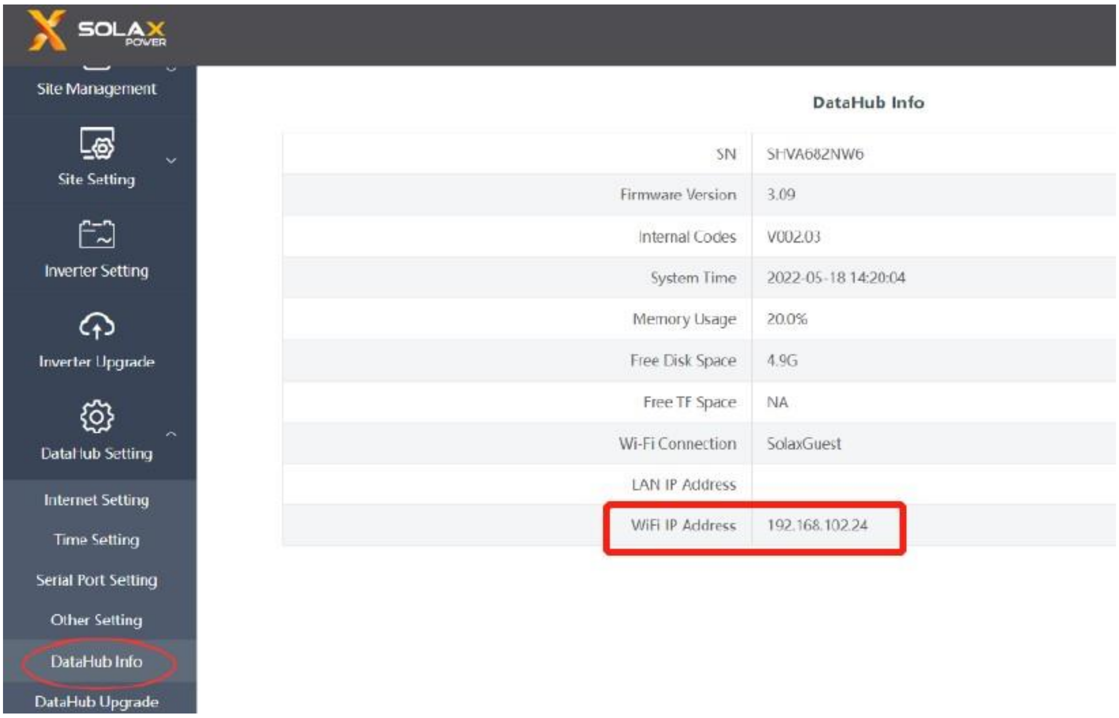
1. For 40600 active power, X3-Hybrid-G4, X3-FIT-G4, X3-Pro-G2, X3-MIC-G2, X3-Ultra, X3-IES, only can be set to 1000 (Only X3-Mega G2, X3-Forth can be set a value to 1100), if you set a value 1100, the Inverter will treat it as 1000 and shows failure on address 45600 and 45601. (In this case, if you use Mega and other Inverters at the same system, and you set a value 1100, the Mega will be set to 1100, and other inverters will be set to 1000)

2. For 40602 Real time active power, X3-Hybrid-G4, X3-FIT-G4, X3-Ultra, X3-IES, don’t have this function currently. (Only X3-Pro-G2, X3-MIC-G2 , X3-Mega G2, X3-Forth has this function) So if you use On-grid Inverter together with Hybrid Inverters, we suggest you to use 40600 to adjust the active power, it has the same function with 40602.

How to Build up Datahub ModbusTCP Connection

1. Follow datahub1000 installation guide to register Solaxcloud monitor account.

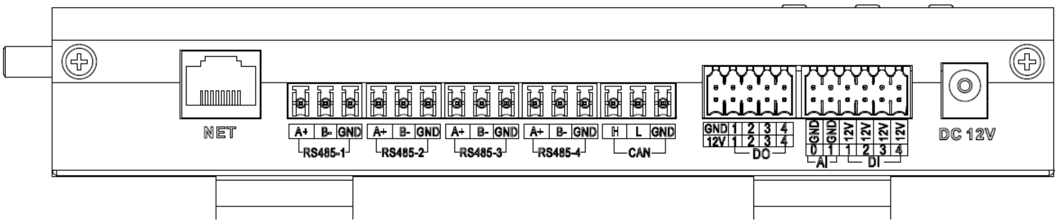
2. Get the datahub IP address. For WiFi connection, get the WiFi IP address, for LAN connection, get the LAN IP address. There are two ways to have the datahub IP address, you can login monitor account and follow below screen shot to check the IP address, or connect to datahub WiFi signal, use local mode to login http://192.168.10.10 to check the IP address. (For the early generation datahub please login http://5.8.8.8, please contact us to confirm the generation).



3. Use PC or laptop to connect to the same home router that datahub connected, open modbus software to input the IP address, start to inquiry the register data.

Data Reading

DataHub has 4 485 ports, each string is connected to a maximum of 20 devices, of which each device has a common read register address of 40, so the total number of registers under each serial port is 40*20 = 800.



Reading RS485 communication address starts from **35000**, so the starting address of the four serial ports is: 35000, 35000+800, 35000+800*2, 35000+800*3, and the modbus address of the device needs to be calculated through the formula.

You can query the address of the parameter you want to know through this formula: ***Start address + (serial port number-1) * 800 + (device address-1) * 40 + offset*** The offset can be viewed by yourself according to the table below.

(This table is for modbus function code 04.)

Number	Name	R/W	Size	Type	Units	Offset address	Discription	byte
1	GridVoltage_LineAB	RO	1	Uint16	0.1V	0	GridLineVoltageAB	2
2	GridVoltage_LineBC	RO	1	Uint16	0.1V	1	GridLineVoltageBC	2
3	GridVoltage_LineCA	RO	1	Uint16	0.1V	2	GridLineVoltageCA	2
4	GridACurrent	RO	1	Int16	0.1A	3	ACsidePhaseACurrent	2
5	GridBCurrent	RO	1	Int16	0.1A	4	ACsidePhaseBCurrent	2
6	GridCCurrent	RO	1	Int16	0.1A	5	ACsidePhaseCCurrent	2
9	GridFrequency	RO	1	Uint32	0.01HZ	6	ACsideGridFrequency	2
7	ActivePower	RO	2	Int32	1W	7	OutputTotalActivePower	4
8	ReactivePower	RO	2	Int32	1Var	9	OutputTotalReactivePower	4
10	PowerFactor	RO	1	Int16	0.001	11	PowerFactor	2

You can also read the data from meter directly, which refer to the table below.

Funcition code is 04								
Using an electricity meter to read parameters from the inverter								
number	Name	R/W	Size	Type	Units	address	Description	byte
1	GridVoltage_LineAB	RO	1	Uint16	0.1V	40440	GridVoltage_LineAB	2
2	GridVoltage_LineBC	RO	1	Uint16	0.1V	40441	GridVoltage_LineBC	2
3	GridVoltage_LineCA	RO	1	Uint16	0.1V	40442	GridVoltage_LineCA	2
4	GridACurrent	RO	1	Int16	0.1A	40443	GridACurrent	2
5	GridBCurrent	RO	1	Int16	0.1A	40444	GridBCurrent	2
6	GridCCurrent	RO	1	Int16	0.1A	40445	GridCCurrent	2
7	ActivePower	RO	2	Int32	1W	40446	ActivePower	4
8	ReactivePower	RO	2	Int32	1Var	40448	ReactivePower	4
9	PowerFactor	RO	1	Int16	0.001	40450	PowerFactor	2
10	GridFrequency	RO	1	Uint16	0.01HZ	40451	GridFrequency	2

For example,

1. Address query

Now I want to know the **start address** of the third device under serial port 2. The starting address should be:

*35000+ (2-1) * 800 + (3-1) *40 = 35880*

(Since each device has a common read register address of 40, the 35880-35919 are all this device' s address. Apparently, the next one' s (the forth device under serial port 2) starting address is 35920.)

2. Parameter Address

   Now I want to know the address of the **AC side grid frequency** of the third device under serial port 2. The address should be:

*35000+ (2-1) * 800 + (3-1) *40 + 6 = 35886*

Except reading from the meter directly, you can also read below Modbus address to get the **Overall active and reactive power** of the Inverter system from Inverter directly. (These values are directly from the Inverter and automatically summed, due to the system self-consumption they may have a bit difference from the Overall active and reactive power you read from the meter.)

Funcition code is 04								
Overall data of the inverter system								
number	Name	R/W	Size	Type	Units	address	Description	byte
1	Overall active power of the inverter	RO	2	Int32	1W	40520	Overall active power of the inverter	4
2	Overall reactive power of the inverter	RO	2	Int32	1Var	40522	Overall reactive power of the inverter	4
3	Power output of the inverter today	RO	1	Uint16	0.01kWh	40524	Daily Yield	2
4	Total power output of the inverter	RO	2	Uint32	1kWh	40526	Total Yield	4
5	Installed capacity	RO	2	Uint32	0.01kW	40528	System Size	4
6	Current power	RO	1	Int16	0.01kW	40530	Output Power	2
7	Carbon dioxide emission reduction	RO	1	Uint16	0.01T	40532	CO2 Reduction	2
8	Benefits & Savings	RO	2	Uint32	1	40534	Income&Saving	4
9	Grid power	RO	2	Int32	0.01kW	40536	Grid Power	4
10	SOC	RO	1	Unit16	1%	40538	SOC	2
11	Grid fault	RO	2	Unit16	/	40539	Grid related Fault Alarm	4

From address 40520 to 40536, you can also read system related information from Datahub, these information are also displayed on the first local page of Datahub.

For Grid related Fault alarm:
This function will send you alarm when the system is reporting grid related fault like grid lost, grid over frequency and so on. You can read the result from 40539 and 40540. If the system is normal, then 40539, 40540 will be both 0; if there are part of the devices reporting grid related fault, then 40539 will be 1, 40540 will be 0; if all the devices are reporting grid related fault, then 40539, 40540 will be both 1.

SOC and device working status information

Funcition code is 04								
Inverter operating status，Read start address40620，Read length is the number of inverters * 2. A maximum of 160 characters are supported								
number	Name	R/W	Size	Type	Units	address	Description	byte
1	RS485 Port and Device number	RO	1	Int16	/	40620	RS485 port number+ device number	2
2	Inverter status	RO	1	Int16	/	40621	RS485 port number+ device working status	2
...	...	...	...	...	...	...	...	...
159	RS485 Port and Device number	RO	1	Int16	/	40798	RS485 port number+ device number	2
160	Inverter status	RO	1	Int16	/	40799	RS485 port number+ device working status	2
161	RS485 Port and Device number	RO	1	Int16	/	40780	RS485 port number+ device number	2
162	Inverter SOC	RO	1	Int16	/	40781	RS485 port number+ device SOC	2
...	...	...	...	...	...	...	...	...
319	RS485 Port and Device number	RO	1	Int16	/	40958	RS485 port number+ device number	2
320	Inverter status	RO	1	Int16	/	40959	RS485 port number+ device SOC	2

SOC:

For 40538 SOC, this is for system average SOC. You can also read the SOC of each device.

This function is 2 address combination,

For example:

40780 will tell you the RS485 port number and device number, 40780 will tell you the RS485 port number and device SOC.
If you read 40780 is 102, it means it’ s the second device of first port, then you read 40781 is 199, it means this device’ s SOC is 99%.

Device working status:

This function is also 2 address combination,

40620 will tell you the RS485 port number and device number, 40621 will tell you the RS485 port number and device working status.

Please refer to below table for working code explanation.

Model	Code	Mode
X3-MIC-G2 / X3-PRO-G2	0	Wait Mode
	1	Check Mode
	2	Normal Mode
	3	Fault Mode
	4	Permanent Fault Mode
	5	Update Mode
X3-FTH / X3-MGA-G2	0	Init Mode
	1	Idle Mode
	2	Start Mode
	3	Run Mode
	4	Fault Mode
	5	Update Mode
X3-Hybrid-G4 / X3-Fit-G4 / X3-ULT	0	Waiting
	1	Checking
	2	Normal
	3	Fault
	4	Permanent Fault
	5	Update
	6	EPS waiting
	7	EPS
	8	Self Testing
	9	Idle
	10	Standby
X3-IES	0	Waiting
	1	Checking
	2	Normal
	3	Fault
	4	Permanent Fault
	5	Update
	6	EPS waiting
	7	EPS
	8	Self Testing
	9	Idle
	10	Standby
	11	Init Mode

Data Writing

At present, DataHub only supports 06 single write, enter the value to the

Function code is 06								
number	**Name**	**R/W**	**Size**	**Type**	**Units**	**Offset address**	**Description**	byte
1	Active power	WO	1	Uint16	0.1%	40600	Active power [0-1100]	2
2	Reactive power cos φ	WO	1	Int16	0.1%	40601	Reactive power cos φ + :Ahead, - :Delay [-1000,-800]&[800,1000]	2
3	Real time active power	WO	1	Uint16	0.1%	40602	Real time active power [0-1100]	2

corresponding offset address. (**Please note:**

1. For 40600 active power, X3-Hybrid-G4, X3-FIT-G4, X3-Pro-G2, X3-MIC-G2, X3-Ultra, X3-IES, only can be set to 1000 (Only X3-Mega G2, X3-Forth can be set a value to 1100), if you set a value 1100, the X3-Hybrid-G4, X3-FIT-G4, X3-Pro-G2, X3-MIC-G2, X3-Ultra, X3-IES, will treat it as 1000 and shows failure on address 45600 and 45601. (In this case, if you use Mega and other Inverters at the same system, and you set a value 1100, the Mega will be set to 1100, and other inverters will be set to 1000)

2. For 40602 Real time active power, X3-Hybrid-G4, X3-FIT-G4, X3-Ultra, X3-IES, don’t have this function currently. (Only X3-Pro-G2, X3-MIC-G2, X3-

Mega G2, X3-Forth has this function) So if you use On-grid Inverter together with Hybrid Inverters, we suggest you to use 40600 (AC port power limitation) to adjust the active power, it has the same function with 40602.

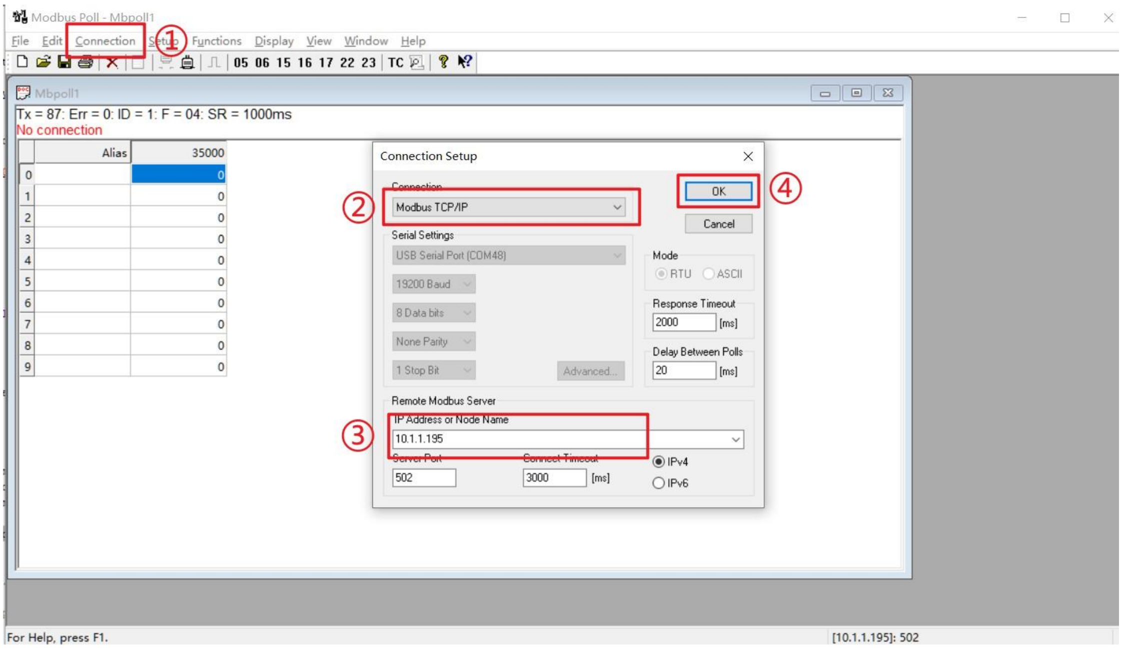
After you finish the writing you can read below address to check if your operation is successfully.

Function code is 04								
Read Write Data								
number	**Name**	**R/W**	**Size**	**Type**	**Units**	**Offset address**	**Description**	byte
1	Active power	RO	1	Uint16	0.1%	45600	Active power [0-1100]	2
2	Write error port	RO	1	Uint16	/	45601	Write error port	2
3	Reactive power cos **φ**	RO	1	Int16	0.1%	45602	Reactive power cos φ + :Ahead, - :Delay [-1000,-800]&[800,1000]	2
4	Write error port	RO	1	Uint16	/	45603	Write error port	2
5	Real time active power	RO	1	Uint16	0.1%	45604	Real time active power [0-1100]	2
6	Write error port	RO	1	Uint16	/	45605	Write error port	2

When a failure occurs, the corresponding address will show a value 2580, and the wirte error port address below it will show the port failed with number 1,2,3,4 (1,2,3,4 is corresponding to port 1,2,3,4)

Software Operation: Use the software Modbus Poll

1、 Connect to Datahub.



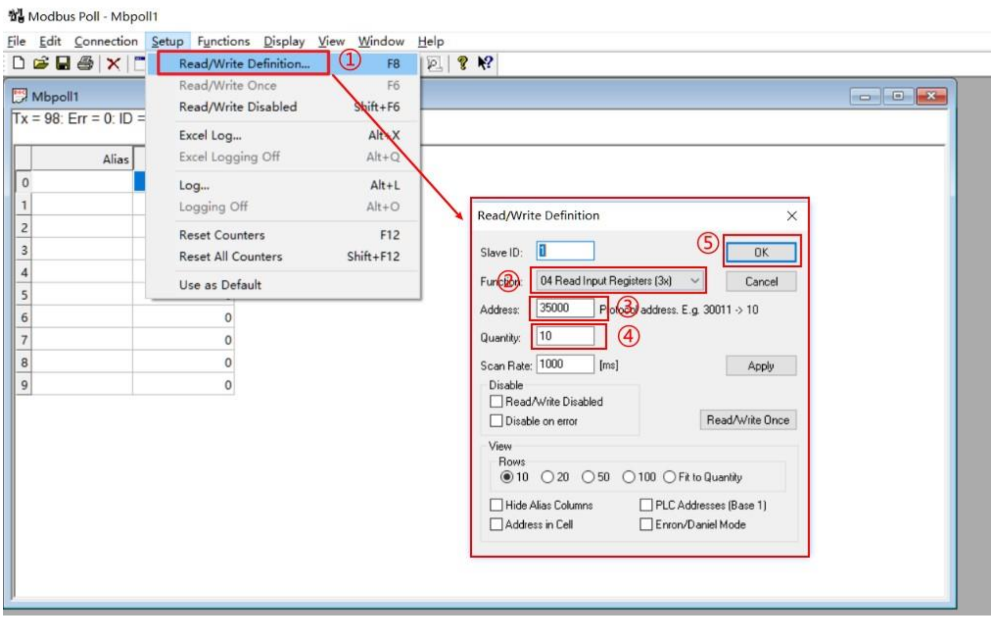
(1) Click Connection, select Connect, and the interface will pop up

(2) Select Modbus TCP/IP

(3) Enter the IP of DataHub

(4) Click OK

2、 Query Inverter Information



(1) Select Read/Write Definition...

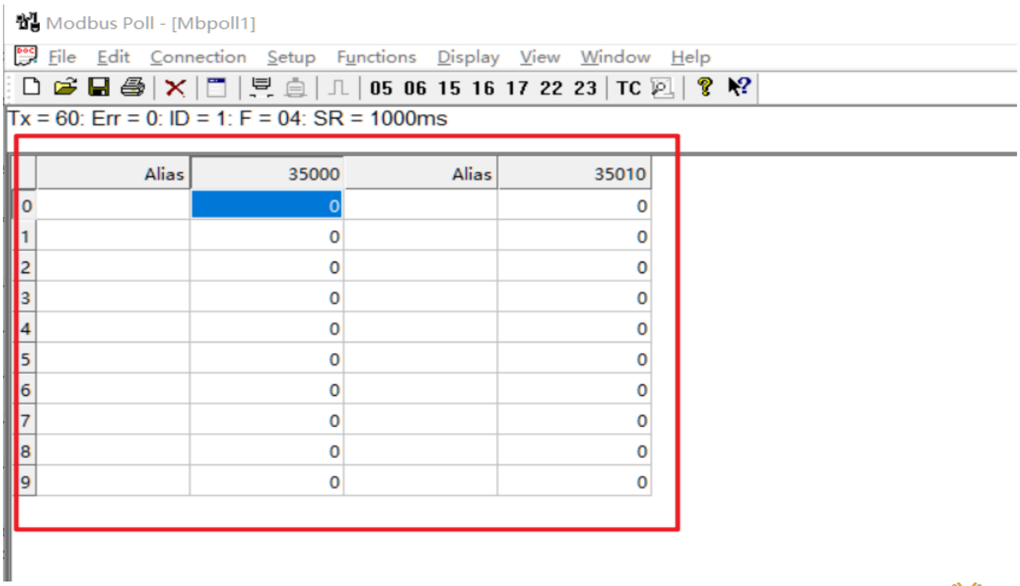
(2) Select 04 Read Input Registers (3x)

(3) Enter the calculated start address of the device you want to query

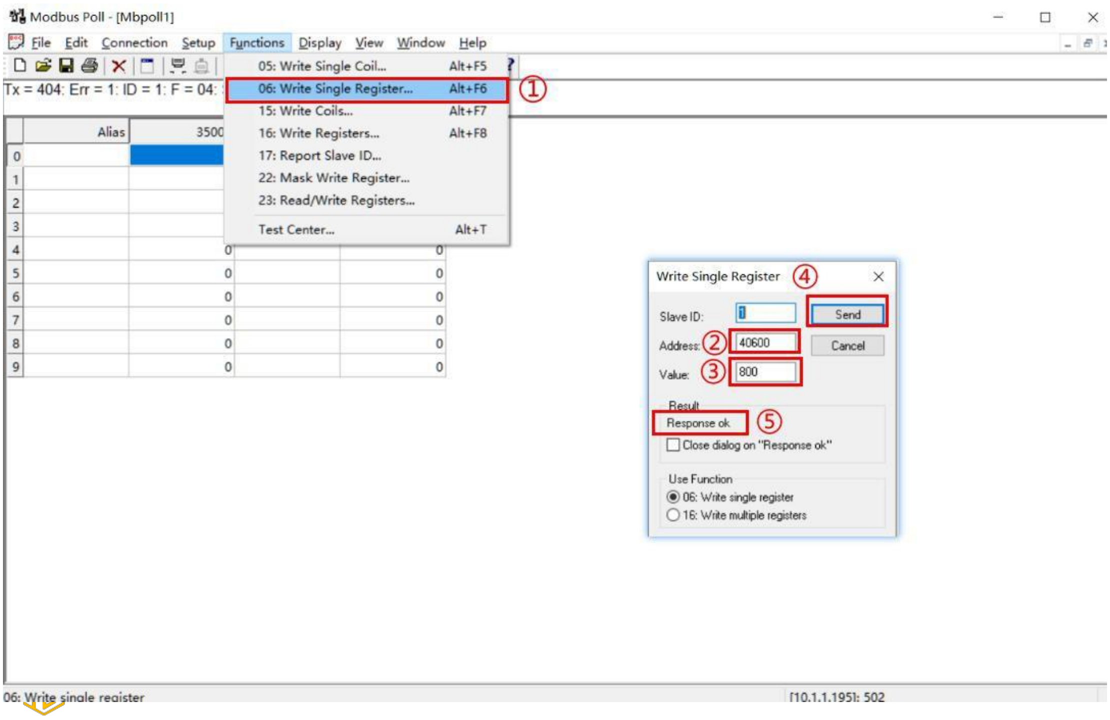
(4) Enter the query length (40 registers per machine, up to 125 registers per query) (Which means you can query no more than 3 device at a time)

(5) Click OK

3、Return Results



4、 Data Write



(1) Select 06: Write Single Register

(2) Enter the register address to be written

(3) Enter the value written by the register address

(4) Click Send

(5) Return Response OK to write successfully.

Compatibility

Device	Firmware
Datahub1000	V017.16（Testing Version） Or V018 (Official Version)

KNOWLEDGE BASE